Department of Information Technology
Academic Year 2023 - 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT
Course Code & Title: CS3452 & Theory of Computation
Name of the Faculty member (s): Dr.V.Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II **/ Equivalence of Finite Automata and regular expressions**
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** Learning by Doing
- **Justification:**
  In unit 2, the students need to understand the Equivalence of Finite Automata and regular expressions. This activity helps the students to understand the voluntarism and leadership.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**
  - A topic Equivalence of Finite Automata and regular expressions was given to Ms. S.Sobana one week before the presentation.
  - He prepared the presentation and taken the class on 01.04.2024.
  - The topic was delivered around 20 minutes with an example problem.
  - The students discussed with an example and gave one question for practicing the method. The students practiced and performed well.
  - The implementation of Learning by doing activity as shown in figure1 and figure2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	2	3	1	2	1	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO8**	**PO9**	**PO10**	**PO12**	**PSO1**	**PSO2**	**PSO3**
	3	2	3	1	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of regular expression to finite automata	The students will be able to identify the need for Finite automata of complex engineering problems	Able to design solutions for any applications using finite automata	Able to analyze and apply finite automata concept	Able to follow ethics for solving the problem	Able to implement the problem individually	Able to implement Equivalence of finite automata and regular expression..	Ability to recognize the need for Finite automaton in real world applications	Exhibit knowledge in finite automata for application development	Exhibit knowledge in finite automata in TOC of IoT, Cloud Computing and Data Analytics to provide IT solutions	Exhibit knowledge in finite automata in software projects to deliver a quality product.

- **Images / Screenshot of the practice:**

Figure 1: Learning by Doing Handled by Sobana S

Figure 2: Learning by Doing

- **Reflective Critique:**
    - ❖ ***Feedback of practice from students and other stakeholders:***

- Every student observed with close attention in class. They are aware of the importance that Equivalence of finite automata is.
- Students understand how regular expressions are equivalent to finite automata.
- It supports students learn about finite automata.

❖ ***Benefit of the practice:***

- Students are capable of using regular expression to analyze finite automata.
- Students can practice NFA to DFA and RE to NFA.
- The significance of finite automata is explained to the students through this activity.

❖ ***Challenges faced in implementation:***

  - Some students did mistakes when following the steps, after clarification the students completed the problem.
  - The topic was planned for 20 minutes, but it took 25 minutes.

References:

❖ https://ohioline.osu.edu/

❖ https://www.ritrjpm.ac.in/images/computer-science/

Signature of Faculty Member **HOD**